Parameter-Free Heavy-Tailed Bandits

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Abstract

Heavy-tailed distributions arise naturally in sequential decision-making problems such as financial investment, online advertising, and network management, where rare but extreme outcomes can dominate performance. Heavy-tailed bandits model online decision-making in these settings by assuming only that rewards X satisfy $\mathbb{E}[|X|^{1+\epsilon}] \leq u$, for some tail exponent $\epsilon \in (0,1]$ and moment bound $u < +\infty$. However, most existing regret minimization algorithms require these parameters to be known. This assumption is particularly restrictive in practice: ϵ and u govern the frequency and magnitude of rare events and are therefore precisely the quantities that are hardest to infer reliably from limited observations.

Motivated by an open problem posed by Genalti and Metelli at COLT 2025, we resolve the assumption-free adaptation problem for heavy-tailed bandits and characterize the price in the regret of not knowing the tail parameters. We first study adaptation to the moment bound u for a fixed tail exponent ϵ. We prove that every algorithm unaware of u, or of any upper bound on it, must obey a sharp trade-off between its distribution-dependent and distribution-free regret guarantees. We then introduce a scheduled-exploration algorithm that requires no knowledge of u and matches the resulting adaptation frontier up to logarithmic factors. Finally, we show that the same algorithm can be instanced without knowing ϵ by calibrating its exploration schedule to the endpoint $\epsilon = 1$. It achieves sublinear regret for every fixed $\epsilon > 0$, while no algorithm can guarantee sublinear regret uniformly over all $\epsilon \in (0,1]$. Altogether, our results resolve the COLT open problem without additional distributional assumptions and provide a sharp characterization of the statistical cost of adapting to unknown heavy tails.

1 Introduction

Heavy-tailed rewards arise naturally in sequential decision-making problems such as financial investment (Gagliolo and Schmidhuber, 2011; Genalti et al., 2026), online advertising (Anderson, 2007), and network management (Liebeherr, Burchard, and Ciucu, 2012), where rare but extreme observations may dominate performance. The heavy-tailed stochastic multi-armed bandit model captures these settings by assuming only that the rewards X of every arm satisfy $\mathbb{E}[|X|^{1+\epsilon}] \leq u$ for some $\epsilon \in (0,1]$ and $u < +\infty$. The parameter ϵ, named *tail exponent*, controls the heaviness of the tails, while u, named *moment bound*, controls their scale. When both parameters are known, robust estimators can be calibrated to attain the distribution-free regret of

$$\widetilde{\mathcal{O}}\big(u^{\frac{1}{1+\epsilon}} K^{\frac{\epsilon}{1+\epsilon}} T^{\frac{1}{1+\epsilon}}\big), \tag{1}$$

where K is the number of arms and T is the horizon (Bubeck, Cesa-Bianchi, and Lugosi, 2013).

The knowledge of (ϵ, u) is particularly restrictive in the real-world. Both parameters describe the behavior of rare observations and are therefore difficult to infer reliably from limited data. Moreover, misspecifying ϵ changes the polynomial concentration rate of the estimators (Lugosi and Mendelson, 2019), rather than merely its constants. Prior work showed that the known-parameter guarantees cannot generally be recovered without either paying an additional regret or imposing further distributional assumptions (Genalti et al., 2024). This motivated the open problem of Genalti and Metelli (2025) asking what are the best assumption-free regret guarantees when the heavy-tail parameters are unknown and which algorithms can attain them.

Contributions. In this paper, we first fix the tail exponent ϵ and study adaptation to the unknown moment bound u. Rather than considering only the best distribution-free rate, we characterize how robustness to arbitrary scales u trades off with performance on favorable instances. Let $\Phi_{free}(K,T)$ denote a moment-free *distribution-free* regret rate and let $\Phi_{dep}(K,T)$ denote the gap-sum-normalized *distribution-dependent* regret rate.[1] We prove that every strategy unaware of u must satisfy (Theorem 4)

$$\Phi_{dep}(K,T)\Phi_{free}(K,T)^{\frac{1+\epsilon}{\epsilon}} = \Omega\big(T^{\frac{1+\epsilon}{\epsilon}}\big). \tag{2}$$

Thus, improving the distribution-free guarantee necessarily deteriorates the distribution-dependent one. This establishes a *frontier* that reveals how adaptation trades off distribution-dependent and distribution-free guarantees.

We complement this lower bound by proposing an adaptive regret minimization algorithm, `Adaptive Robust ETC` (`AdaR-ETC`), which leverages Median-of-Means (Lugosi and Mendelson, 2019) Explore-Then-Commit (Lattimore and Szepesvári, 2020) strategy that does not make use of the knowledge of u. To achieve adaptivity, `AdaR-ETC` is parametrized by $\alpha \in [(1+\epsilon)/(1+2\epsilon), 1)$, $q \in [0, \epsilon/(1+2\epsilon)]$, and $\beta_\alpha = (1-\alpha)(1+\epsilon)/\epsilon$, and obtains a distribution-free regret bound (Theorem 5),

$$\Phi_{free}(K,T) = \widetilde{\mathcal{O}}\big(K^{\frac{\epsilon}{1+\epsilon}(1-q)}T^{\alpha}\big) \tag{3}$$

and a distribution-dependent regret bound (Theorem 6),

$$\Phi_{dep}(K,T) = \widetilde{\mathcal{O}}\big(K^{q-1}T^{\beta_\alpha}\big). \tag{4}$$

The combination of these guarantees is tight on the joint lower bound frontier of Equation (2). Balancing the horizon dependence and the one on the number of arms gives

$$\widetilde{\mathcal{O}}\big(u^{\frac{1}{1+\epsilon}}K^{\frac{\epsilon}{1+2\epsilon}}T^{\frac{1+\epsilon}{1+2\epsilon}}\big). \tag{5}$$

This also represents the best possible distribution-free guarantee that can be obtained by any algorithm unaware of u. As visible from the exponent of T, this regret bound is worse compared to the one of Equation (1), establishing the price of adaptivity. Specializing this bound in the finite variance case ($\epsilon = 1$), we obtain a $\widetilde{\mathcal{O}}(T^{\frac{2}{3}})$ rate, strictly greater than the $\widetilde{\mathcal{O}}(\sqrt{T})$ rate attainable in bandits with bounded or subgaussian rewards (Lattimore and Szepesvári, 2020).

We then remove the knowledge of ϵ. The Median-of-Means estimator uses neither ϵ nor u; only the exploration schedule makes use of ϵ. Calibrating it to the known endpoint $\epsilon = 1$ yields a single strategy, independent of both parameters, satisfying a distribution-free regret bound

$$\widetilde{\mathcal{O}}\big(u^{\frac{1}{1+\epsilon}}K^{\frac{2\epsilon}{3(1+\epsilon)}}T^{\frac{3+\epsilon}{3(1+\epsilon)}}\big) \tag{6}$$

for every fixed true $\epsilon > 0$ (Theorem 7). Thus matches the bound of Eq. (5) for $\epsilon = 1$, but deteriorates for all $\epsilon \in (0,1)$. Furthermore, we prove a pairwise lower bound (Theorem 8) across moment orders showing that this profile is optimal, up to logarithmic factors, among strategies retaining the balanced finite-variance guarantee at $\epsilon = 1$. Hence, no single policy is optimal at every moment order; adaptation is described by a frontier rather than by one oracle curve.

Finally, our pointwise guarantee cannot be made uniform. Although the regret is sublinear for every fixed $\epsilon > 0$, no strategy can guarantee sublinear regret (normalized by u) uniformly over $\epsilon \in (0,1]$ (Corollary 9). Indeed, as ϵ approaches zero, the finite-moment assumption becomes arbitrarily weak and the exponent of T approaches one.

Altogether, our results characterize the assumption-free cost of u-adaptivity, provide an algorithm attaining the resulting distribution-dependent frontier, and identify the limits of simultaneous (ϵ, u)-adaptation.

2 Heavy-Tailed Bandits

We recall some fundamental notions on heavy-tailed bandits (Bubeck, Cesa-Bianchi, and Lugosi, 2013) and the required notations for regret rates defined in (Hadiji and Stoltz, 2020).

[1] These quantities will be formally defined later in the paper.

Interaction Protocol. In the stochastic multi-armed bandit problem (Lattimore and Szepesvári, 2020), a learner interacts with $K \in \mathbb{N}_{\geq 2}$ arms for a horizon of $T \in \mathbb{N}$ rounds. A bandit instance is an ordered tuple $\underline{\nu} = (\nu_1, \dots, \nu_K)$ of probability distributions on $\mathbb{R}$. For every arm $i \in [K]$,[2] successive pulls produce an i.i.d. sequence $X_{i,1}, X_{i,2}, \dots \overset{\text{i.i.d.}}{\sim} \nu_i$, and the reward sequences are independent across arms.

At every round $t \in [T]$, the learner selects an arm $I_t \in [K]$ according to a possibly randomized rule π measurable w.r.t. the history up to $t-1$. The learner then observes the reward generated by the selected arm $X_{I_t,t}$. Let $N_i(t) = \sum_{s=1}^{t} \mathbf{1}\{I_s = i\}$ be the number of pulls of arm i up to t.

Heavy-Tailed Bandits. In this paper, we consider heavy-tailed reward distributions. For $\epsilon \in (0,1]$ and $u > 0$, we denote the set of heavy-tailed bandit instances as:

$$\mathcal{H}_{\epsilon,u} = \left\{ (\nu_1, \dots, \nu_K) : \mathbb{E}_{X \sim \nu_i}\left[|X|^{1+\epsilon}\right] \leq u, \ \forall i \in [K] \right\}.$$

The dependence of $\mathcal{H}_{\epsilon,u}$ on K is kept implicit in the notation.

Let $\mu_i \coloneqq \mathbb{E}_{X \sim \nu_i}[X]$ be the expected reward of arm i, $\mu^* \coloneqq \max_{i \in [K]} \mu_i$ be the optimal expected reward, and $\Delta_i \coloneqq \mu^* - \mu_i$ be the suboptimality gap of arm i. The expected cumulative regret of a strategy π is given by

$$R_T^{\pi}(\underline{\nu}) = \mathbb{E}_{\underline{\nu},\pi}\left[\sum_{t=1}^{T} (\mu^* - \mu_{I_t})\right] = \sum_{i=1}^{K} \Delta_i \mathbb{E}_{\underline{\nu},\pi}[N_i(T)],$$

where the expectation is taken over both the randomness rewards and the internal randomness of the strategy. When the strategy is clear from the context, we write $R_T(\underline{\nu})$.

(ϵ, u)-adaptivity in Heavy-Tailed Bandits. Most of the existing algorithms require, as an input, both u and ϵ (see, e.g., Bubeck, Cesa-Bianchi, and Lugosi (2013); Agrawal, Juneja, and Glynn (2020); Lee and Lim (2022)). These parameters govern the behavior of the tails of the reward distributions and cannot be estimated reliably (Bahadur and Savage, 1956). Since heavy-tailed bandits model complex real-world scenarios beyond the canonical yet limiting distributional assumptions, requiring such knowledge severely limits their scope. A recent research stream (Ashutosh et al., 2021; Genalti et al., 2024; Tamás, Szentpéteri, and Csáji, 2024; Chen et al., 2024) focused on devising (ϵ, u)-adaptive algorithms, *i.e.*, unaware of the values of u and/or ϵ, and on characterizing the statistical limits of learnability without this knowledge. Genalti et al. (2024) show that adaptivity comes at a cost: either that knowledge is substituted by another structural assumption, or the same regret bounds as if u and/or ϵ are known (*oracle rates*) cannot be achieved.

Genalti and Metelli (2025) propose a COLT open problem addressing the following questions:

1. *What are the best possible regret rates that can be achieved under adaptivity requirements?*
2. *What algorithms match such rates?*
3. *Is there a best assumption that allows for oracle rates?*

In this paper, we provide a collection of results that, altogether, answer the first two questions.

3 Regret Rates in Heavy-Tailed Bandits

In the stochastic bandit literature, there are two main ways to express regret guarantees: *distribution-free* bounds and *distribution-dependent* bounds.[3] In this paper, the latter term refers specifically to the gap-sum-normalized coefficient introduced in Definition 3. In distribution-dependent bounds, the guarantee depends on the specific instance through the suboptimality gaps Δ_i, whereas distribution-free bounds remove this dependence by considering the worst case over the entire class. We recall the known lower bound which characterizes the minimax regret in heavy-tailed bandits when both u and ϵ are known to the learner.

Theorem 1 (Distribution-free regret lower bound, Bubeck, Cesa-Bianchi, and Lugosi (2013))**.** *Fix $\epsilon \in (0,1]$. There exists a constant $c_\epsilon > 0$, depending only on ϵ, such that, for every $u > 0$, every $K \geq 2$, every horizon*

[2] Given $k \in \mathbb{N}$, we define $[k] \coloneqq \{1, 2, \dots, k\}$.

[3] With little approximation, these guarantees are also known in the literature as *worst-case* and *instance-dependent*.

$T \geq K$, *and every exploration strategy* π,

$$\sup_{\underline{\nu} \in \mathcal{H}_{\epsilon,u}} R_T^\pi(\underline{\nu}) \geq c_\epsilon u^{\frac{1}{1+\epsilon}} K^{\frac{\epsilon}{1+\epsilon}} T^{\frac{1}{1+\epsilon}}. \tag{7}$$

In this paper, we tackle (ϵ, u)-adaptivity through a two-step approach. First, we characterize the u-adaptive setting, in which ϵ is known. We then remove the knowledge of ϵ and quantify the additional difficulty of simultaneously adapting to both parameters. It is worth noting that the he u-adaptive setting is of interest on its own. Indeed, in non-heavy-tailed MABs, adaptation to an unknown bound on the support of the rewards has been characterized in Hadiji and Stoltz (2020). However, in heavy-tailed bandits the limits of adaptation to an unknown moment bound remain open.

Inspired by the definitions of Hadiji and Stoltz (2020) for bounded-support bandits, we now define the two types of regret rates considered in our analysis.

Definition 2 (Moment-free distribution-free regret rate). *A strategy* π *for stochastic heavy-tailed bandits admits a moment-free distribution-free regret bound* Φ_{free} *if, without knowing* u*, it guarantees*

$$R_T^\pi(\underline{\nu}) \leq u^{\frac{1}{1+\epsilon}} \Phi_{free}(K, T), \tag{8}$$

for all $K \geq 2$*,* $T \geq 1$*,* $u > 0$*, and* $\underline{\nu} \in \mathcal{H}_{\epsilon,u}$.

Theorem 1 implies that every attainable moment-free distribution-free rate, whenever $T \geq K$, satisfies

$$\Phi_{free}(K, T) \geq c_{\mathrm{std},\epsilon} K^{\frac{\epsilon}{1+\epsilon}} T^{\frac{1}{1+\epsilon}}, \tag{9}$$

for some constant $c_{\mathrm{std},\epsilon} > 0$ possibly depending on ϵ.

Definition 3 (Distribution-dependent regret rate). *A strategy* π *for stochastic heavy-tailed bandits admits a distribution-dependent rate* Φ_{dep} *if, without knowing* u*, it guarantees*

$$\limsup_{T \to +\infty} \frac{R_T^\pi(\underline{\nu})}{\Phi_{dep}(K, T)} \leq \sum_{i : \Delta_i > 0} \Delta_i, \tag{10}$$

for all $K \geq 2$*,* $u > 0$*,* $\underline{\nu} \in \mathcal{H}_{\epsilon,u}$.

The *normalization* in Definition 3 entails no loss of generality. Indeed, any multiplicative constant in a distribution-dependent upper bound can be absorbed into the definition of $\Phi_{dep}(K, T)$. This is necessary to compare rates and to state the adaptation frontier with a constant that depends only on ϵ. The term distribution-dependent rate has a specific meaning in this paper: $\Phi_{dep}(K, T)$ is the coefficient multiplying the sum of the suboptimality gaps. It should not be confused with the classical distribution-dependent bounds, which may exhibit a different dependence on the gaps.

The functions $\Phi_{free}(K, T)$ and $\Phi_{dep}(K, T)$ explicitly depend on both the horizon T and the number of arms K. Their dependence on ϵ is suppressed because ϵ is fixed throughout the u-adaptive analysis, whereas neither rate is allowed to depend on the unknown value of u. In the following sections, we characterize the trade-off between these two rates and study their optimal dependence on K, T, and ϵ.

4 u-adaptivity: *You can't have it both ways*

In this section, we characterize the limits of learnability when the moment bound u is unknown. Our main result establishes a fundamental trade-off between the distribution-free and the distribution-dependent guarantees. These two guarantees cannot be optimized independently: improving one necessarily deteriorates the other. Moreover, the trade-off concerns both the dependence on the horizon T and the dependence on the number of arms K. The following result shows that the two quantities must lie on a frontier.

Theorem 4 (Existence of a trade-off). *Fix* $\epsilon \in (0, 1]$*. Consider a strategy that does not know* u *and admits a moment-free distribution-free rate* $\Phi_{free}(K, T) = o(T)$*, for every fixed* $K \geq 2$*. Then, any distribution-*

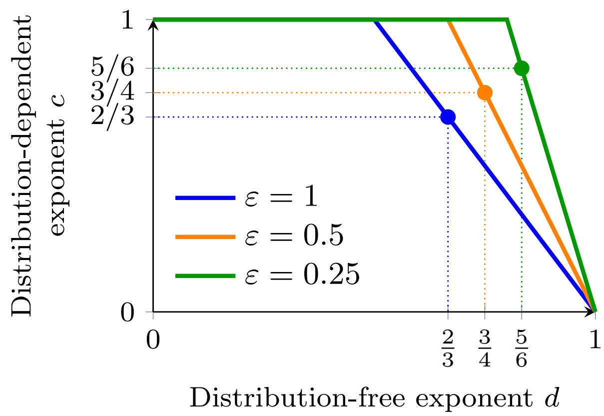

Figure 1: Trade-off between the distribution-dependent and the distribution-free rates in T.

dependent rate $\Phi_{dep}(K,T)$ satisfying Definition 3 fulfills

$$\liminf_{T\to+\infty} \frac{\Phi_{dep}(K,T)\Phi_{free}(K,T)^{\frac{1+\epsilon}{\epsilon}}}{T^{\frac{1+\epsilon}{\epsilon}}} \geq c_\epsilon, \quad \forall K \geq 2, \tag{11}$$

where $c_\epsilon > 0$ depends only on ϵ.

The proof follows the change-of-measure procedure developed in (Hadiji and Stoltz, 2020), together with the instance construction of Genalti et al. (2024).

Theorem 4 establishes a frontier rather than two independent lower bounds. Intuitively, a learner that aggressively pursues a small distribution-dependent regret explores apparently suboptimal arms only a limited number of times. In the heavy-tailed setting, however, an arm that mostly returns low-reward observations may still hide a rare but extremely large reward. Protecting against these alternatives requires additional exploration. This improves the distribution-free guarantee, but it is unnecessary on favorable instances and deteriorates the distribution-dependent performance.

The tension first appears in the dependence on T and, among strategies lying on the optimal horizon frontier, also in the dependence on K. To isolate the exponents, suppose that the two rates admit monomial envelopes of the form $\Phi_{dep}(K,T) = K^aT^c$ and $\Phi_{free}(K,T) = K^bT^d$, up to multiplicative factors that are bounded above and below by constants independent of K and T. Substituting these expressions into Theorem 4 gives

$$\liminf_{T\to+\infty} K^{a+\frac{1+\epsilon}{\epsilon}b}T^{c+\frac{1+\epsilon}{\epsilon}d-\frac{1+\epsilon}{\epsilon}} \geq c_\epsilon.$$

Consequently, the exponents of T must satisfy $c + d(1+\epsilon)/\epsilon \geq (1+\epsilon)/\epsilon$. This inequality describes the fundamental trade-off in the horizon T. Decreasing the distribution-free exponent d forces the distribution-dependent exponent c to increase, and vice versa. In Figure 1, we provide a graphical representation of this trade-off.

The dependence on K requires additional care because Theorem 4 takes T to infinity for each fixed K. If the inequality holds strictly, polynomial growth in T may compensate for any fixed dependence on K. Consider instead strategies attaining the horizon boundary $c + d(1+\epsilon)/\epsilon = (1+\epsilon)/\epsilon$. For these strategies, the dependence on T cancels in the lower bound. Since c_ϵ is independent of K, the exponent of K must then satisfy $a + b(1+\epsilon)/\epsilon \geq 0$. Thus, among strategies lying on the optimal horizon frontier, improving the distribution-free dependence on K necessarily deteriorates the distribution-dependent dependence.

Equal-gap interpretation. Consider an equal-gap instance consisting of one optimal arm and $K-1$ suboptimal arms, each with gap $\Delta > 0$. On this family, $\sum_{i:\Delta_i>0}\Delta_i = (K-1)\Delta$. Therefore, if $\Phi_{dep}(K,T) = K^aT^c$, then

$$R_T(\underline{\nu}) \leq K^aT^c(K-1)\Delta \leq K^{a+1}T^c\Delta.$$

Thus, the exponent governing the dependence of the actual regret on K is $a+1$, rather than a. Rewriting the inequality in terms of the distribution-dependent exponent gives $(a+1) + b(1+\epsilon)/\epsilon \geq 1$. This formulation

Table 1: Representative points on the K-frontier for $\epsilon = 1$.

Operating point	(b, a)	$\Phi_{free}(K, T)$	Equal-gap R_T
Instance-oriented	$(1/2, -1)$	$K^{1/2}T^d$	$\mathcal{O}(T^c\Delta)$
K-balanced	$(1/3, -2/3)$	$K^{1/3}T^d$	$\mathcal{O}(K^{1/3}T^c\Delta)$

clarifies that both the distribution-free regret and the actual distribution-dependent regret may deteriorate as K increases. The tension is not that one quantity must decrease while the other increases. Rather, their exponents in K cannot both be made arbitrarily small. On the boundary of the K-frontier, we have $a = -b(1+\epsilon)/\epsilon$, and hence the distribution-dependent exponent is $a+1 = 1-b(1+\epsilon)/\epsilon$. Therefore, reducing the distribution-free exponent b necessarily increases the distribution-dependent one $a+1$.

Representative points for $\epsilon = 1$. At the finite-variance endpoint, the K-frontier becomes $a + 2b \geq 0$. Table 1 reports two representative points on its boundary. The first choice yields a distribution-dependent regret that is essentially independent of K on the equal-gap family, but pays a $K^{1/2}$ distribution-free factor. Moving to the second point improves the distribution-free dependence from $K^{1/2}$ to $K^{1/3}$, while the equal-gap distribution-dependent regret deteriorates from a constant dependence on K to $K^{1/3}$.

Balancing both K and T. A natural operating point is obtained by requiring the two guarantees to have the same polynomial dependence on both the horizon and the number of arms. For the horizon, imposing $c = d$ gives $c = d \geq (1+\epsilon)/(1+2\epsilon)$. Thus, when the two guarantees are required to have the same dependence on T, neither exponent can be smaller than $(1+\epsilon)/(1+2\epsilon)$. For the number of arms, the distribution-free exponent is b, whereas the distribution-dependent exponent on the equal-gap family is $a+1$. Balancing them amounts to imposing $b = a+1$. Combining this identity with the boundary condition $a+b(1+\epsilon)/\epsilon = 0$ gives $b = \frac{\epsilon}{1+2\epsilon}$ and $a = -\frac{1+\epsilon}{1+2\epsilon}$. At the point balancing both K and T, the two rates have form $\Phi_{free}(K,T) = K^{\frac{\epsilon}{1+2\epsilon}}T^{\frac{1+\epsilon}{1+2\epsilon}}$ and $\Phi_{dep}(K,T) = K^{-\frac{1+\epsilon}{1+2\epsilon}}T^{\frac{1+\epsilon}{1+2\epsilon}}$. On the equal-gap family, this corresponds to $R_T(\underline{\nu}) = \mathcal{O}\big(K^{\frac{\epsilon}{1+2\epsilon}}T^{\frac{1+\epsilon}{1+2\epsilon}}\Delta\big)$, which has the same dependence on K and T as the distribution-free rate.

For $\epsilon = 1$, the balanced frontier point is $\Phi_{free}(K,T) = K^{1/3}T^{2/3}$ and $\Phi_{dep}(K,T) = K^{-2/3}T^{2/3}$. In particular, the balanced horizon dependence is $T^{2/3}$, which is worse than the $\sqrt{T}$ dependence arising in adaptation to an unknown bounded reward range (Hadiji and Stoltz, 2020). This deterioration reflects the additional difficulty of ruling out rare and arbitrarily large rewards, typical of heavy-tailed distributions, when only a finite, unknown moment bound is available.

5 Explore-Then-Commit suffices for u-adaptivity

In this section, we propose an algorithm, fully unaware of u, that achieves regret guarantees that are tight on the frontier defined by Theorem 4. The algorithm allows us to select a point on both the T-frontier and the K-frontier. The dependence on T is controlled by the parameter α, whereas the one on K is controlled by an additional exploration parameter q.

Surprisingly, the algorithm is very simple and natural. In fact, an Explore-Then-Commit (ETC) strategy with robust estimation and a tuned amount of exploration L_T is enough to get there. We call our algorithm `Adaptive Robust ETC` (`AdaR-ETC`, for short), and we report its pseudocode in Algorithm 1. In the next paragraphs, we describe the main components of `AdaR-ETC`.

Robust Estimator. Since the distributions are heavy-tailed, the empirical mean is not a suitable estimator (Bubeck, Cesa-Bianchi, and Lugosi, 2013). We then resort to the well-known Median of Means estimator (MoM, for short). Let $\mathcal{F}_i = \{X_{i,1}, \ldots, X_{i,n_i}\}$ be the set of n_i exploration samples collected from arm i. We divide these samples into B_T blocks of equal size $s_i = \lfloor n_i/B_T \rfloor$. For every $b \in [B_T]$, we define the b-th block as $\mathcal{G}_{i,b} = \big\{X_{i,(b-1)s_i+1}, \ldots, X_{i,bs_i}\big\}$. Thus, each block contains exactly s_i samples. If n_i is not divisible by B_T, the remaining $n_i - B_T s_i$ samples are discarded.

Algorithm 1: `Adaptive Robust ETC (AdaR-ETC)`

Require: Number of arms K, horizon T, exploration parameters $\alpha \in [(1+\epsilon)/(1+2\epsilon), 1)$ and $q \in [0, \epsilon/(1+2\epsilon)]$.

```
Set β_α ← (1−α)(1+ε)/ε, B_T ← ⌈8 log(KT^3)⌉, L̃_T ← K B_T + ⌈K^q T^{β_α}⌉, L_T ← min{T, L̃_T}, F_i ← ∅ for all i ∈ [K].
for t = 1, ..., L_T do
    Select arm I_t ← 1 + ((t − 1) mod K).
    Observe the reward and append it to F_{I_t}.
end for
if L_T < T then
    for i ∈ [K] do
        Compute μ̂_i^{MoM}(F_i).
    end for
    Select Î* ∈ arg max_{i∈[K]} μ̂_i^{MoM}(F_i).
    for t = L_T + 1, ..., T do
        Select arm I_t ← Î*.
    end for
end if
```

For each block $\mathcal{G}_{i,b}$, we define the corresponding block average as $\overline{X}_{i,b} = \frac{1}{s_i}\sum_{\ell=(b-1)s_i+1}^{bs_i} X_{i,\ell}$ for $b \in [B_T]$. Let $\overline{X}_{i,(1)} \le \overline{X}_{i,(2)} \le \cdots \le \overline{X}_{i,(B_T)}$ denote the ordered block averages. The MoM estimator is defined as

$$\widehat{\mu}_i^{MoM}(\mathcal{F}_i) = \overline{X}_{i,(\lceil B_T/2 \rceil)}.$$

Intuitively, although a single block average may be corrupted by an extreme observation, under the finite $(1+\epsilon)$-moment assumption, a constant fraction of the block averages remains close to the true mean with high probability. Taking their median prevents a small number of atypical blocks from significantly affecting the estimate.

More precisely, if $\mathbb{E}[|X|^{1+\epsilon}] \le u$, the estimation error is, with high probability, of order $u^{\frac{1}{1+\epsilon}} B_T^{\frac{\epsilon}{1+\epsilon}} n_i^{-\frac{\epsilon}{1+\epsilon}}$. Thus, we choose B_T logarithmic in K and T. Most importantly, while u and ϵ determine the rate appearing in the concentration analysis, the computation of the MoM estimator itself does not require knowledge of u nor ϵ. This estimator enjoys optimal, up to constants, concentration properties around the true mean (Bubeck, Cesa-Bianchi, and Lugosi, 2013).

Exploration Budget. The exploration budget of `AdaR-ETC` is controlled by two parameters. The parameter α determines how the exploration budget scales with the horizon, through $\beta_\alpha = (1-\alpha)(1+\epsilon)/\epsilon$. A larger α corresponds to a smaller β_α and thus to less exploration as T grows. This improves the distribution-dependent rate on T, at the cost of a worse distribution-free rate. The parameter q plays the analogous role for the dependence on the number of arms. The polynomial part of the total exploration budget is $K^q T^{\beta_\alpha}$. Since exploration is performed in a round-robin fashion, each arm gets approximately $K^{q-1}T^{\beta_\alpha}$ samples. Increasing q assigns more exploration samples to each arm as K grows. This improves the distribution-free rate on K, but increases the regret paid on favorable instances.

The restrictions on α and q are chosen so that the exploration contribution does not dominate the estimation one. Indeed, $\alpha \ge \beta_\alpha$ is equivalent to $\alpha \ge (1+\epsilon)/(1+2\epsilon)$, whereas $q \le \epsilon/(1+2\epsilon)$ is equivalent to $q \le (1-q)\epsilon/(1+\epsilon)$.

Regret Guarantees. The following results formalize the resulting trade-off and certify the tightness of `AdaR-ETC` with respect to the frontier of Theorem 4.

Theorem 5 (Distribution-free regret of `AdaR-ETC`)**.** *Let $\epsilon \in (0,1]$ be fixed and known. Let $\alpha \in [(1+\epsilon)/(1+2\epsilon), 1)$ and $q \in [0, \epsilon/(1+2\epsilon)]$. For every $u > 0$ and every instance $\underline{\nu} \in \mathcal{H}_{\epsilon,u}$, `AdaR-ETC` satisfies*

$$R_T^{\texttt{AdaR-ETC}}(\underline{\nu}) \le \widetilde{\mathcal{O}}\big(u^{\frac{1}{1+\epsilon}} K^{\frac{\epsilon}{1+\epsilon}(1-q)} T^{\alpha}\big),$$

where $\widetilde{\mathcal{O}}$ hides polylogarithmic terms in T and constants depending only on ϵ, α, and q.

The two main contributions to the regret are, up to logarithmic factors, $u^{\frac{1}{1+\epsilon}} K^q T^{\beta_\alpha}$ due to exploration, and $u^{\frac{1}{1+\epsilon}} K^{\frac{\epsilon}{1+\epsilon}(1-q)} T^{\alpha}$ due to committing according to the MoM estimates. The restrictions imposed on

α and q ensure that the latter term dominates. Thus, `AdaR-ETC` is u-adaptive with distribution-free rate $\Phi_{free}(K,T) = \widetilde{\mathcal{O}}\big(K^{\frac{\epsilon}{1+\epsilon}(1-q)}T^{\alpha}\big)$.

On the other hand, we have the following distribution-dependent guarantee.

Theorem 6 (Distribution-dependent regret of `AdaR-ETC`)**.** *Let $\epsilon \in (0,1]$ be fixed and known. Let $\alpha \in [(1+\epsilon)/(1+2\epsilon), 1)$ and $q \in [0, \epsilon/(1+2\epsilon)]$. For every fixed instance $\underline{\nu} \in \mathcal{H}_{\epsilon,u}$, `AdaR-ETC` satisfies*

$$\limsup_{T\to+\infty} \frac{R_T^{\texttt{AdaR-ETC}}(\underline{\nu})}{K^{q-1}T^{\beta_\alpha}} \leq \sum_{i:\Delta_i>0} \Delta_i.$$

Thus, `AdaR-ETC` is u-adaptive with distribution-dependent rate $\Phi_{dep}(K,T) = \widetilde{\mathcal{O}}\big(K^{q-1}T^{\beta_\alpha}\big)$.

The two parameters α and q control two distinct, but parallel, trade-offs. The parameter α determines the trade-off in the horizon T: $\Phi_{free}(K,T) \propto T^{\alpha}$ and $\Phi_{dep}(K,T) \propto T^{\beta_\alpha}$. Since $\alpha + \beta_\alpha\epsilon/(1+\epsilon) = 1$, the two exponents lie exactly on the T-frontier. Similarly, the parameter q determines the trade-off in the number of arms K: $\Phi_{free}(K,T) \propto K^{\frac{\epsilon}{1+\epsilon}(1-q)}$ and $\Phi_{dep}(K,T) \propto K^{q-1}$. These exponents satisfy $(1-q)\epsilon/(1+\epsilon) + (q-1)\epsilon/(1+\epsilon) = 0$. Thus, at the level of exponents, the choice of q realizes the equality case of the K-trade-off associated with the optimal T-frontier.

Hence, for every admissible choice of α and q, `AdaR-ETC` matches the T-frontier of Theorem 4 up to logarithmic factors. Its explicit dependence on K realizes the corresponding polynomial K-trade-off on the horizon-optimal boundary.

6 Characterizing (ϵ, u)-adaptivity

We now remove the knowledge of ϵ and consider a single strategy that uses neither u nor ϵ. This setting involves two distinct adaptation constraints. The first is the frontier associated with the unknown scale u, characterized in Theorem 4. The second is a new frontier across different moment orders ϵ: improving the regret guarantee on a lighter-tailed class necessarily worsens the guarantee on heavier-tailed classes.

(ϵ, u)-adaptive `AdaR-ETC`. We first construct an order-free version of `AdaR-ETC` by calibrating both its T-dependence and its K-dependence to the finite-variance endpoint $\epsilon = 1$. We then show that, for every fixed $\epsilon > 0$, the resulting distribution-free and distribution-dependent guarantees lie on the unknown-u frontier. Finally, we prove that its distribution-free guarantee is also tight, up to logarithmic factors, among strategies retaining the optimal endpoint guarantee at $\epsilon = 1$. At the finite-variance endpoint $\epsilon = 1$, the choice balancing the dependence on the horizon T is $\alpha = 2/3$ and $\beta_\alpha = 2/3$, whereas the choice balancing the distribution-free and the distribution-dependent dependence on the number of arms K is $q = 1/3$. Thus, we define the order-free version of `AdaR-ETC` by setting directly $B_T = \lceil 8\log(KT^3)\rceil$ and $\widetilde{L}_T = KB_T + \lceil K^{1/3}T^{2/3}\rceil$. As in the previous section, the effective exploration budget is $L_T = \min\{T, \widetilde{L}_T\}$ and the arms are explored in round-robin order. The resulting strategy is fully unaware of both u and ϵ.

The following theorem characterizes its regret.

Theorem 7 (Regret of `AdaR-ETC` calibrated with $\epsilon = 1$)**.** *Let $\epsilon \in (0,1]$ and $u > 0$ be fixed. For every $T \geq K \geq 2$ and every instance $\underline{\nu} \in \mathcal{H}_{\epsilon,u}$, the order-free version of `AdaR-ETC` calibrated with $\epsilon = 1$ satisfies*

$$R_T^{\texttt{AdaR-ETC}}(\underline{\nu}) \leq \widetilde{\mathcal{O}}\left(u^{\frac{1}{1+\epsilon}}K^{\frac{2\epsilon}{3(1+\epsilon)}}T^{\frac{3+\epsilon}{3(1+\epsilon)}}\right). \tag{12}$$

where $\widetilde{\mathcal{O}}$ hides factors at most polylogarithmic in K and T and constants depending on ϵ. Moreover, for every fixed instance $\underline{\nu} \in \mathcal{H}_{\epsilon,u}$,

$$\limsup_{T\to+\infty} \frac{R_T^{\texttt{AdaR-ETC}}(\underline{\nu})}{K^{-2/3}T^{2/3}} \leq \sum_{i:\Delta_i>0} \Delta_i. \tag{13}$$

The distribution-free guarantee follows from the same exploration–estimation decomposition of Theorem 5. For every fixed K, the regret is sublinear on every fixed heavy-tailed moment class, even though the algorithm

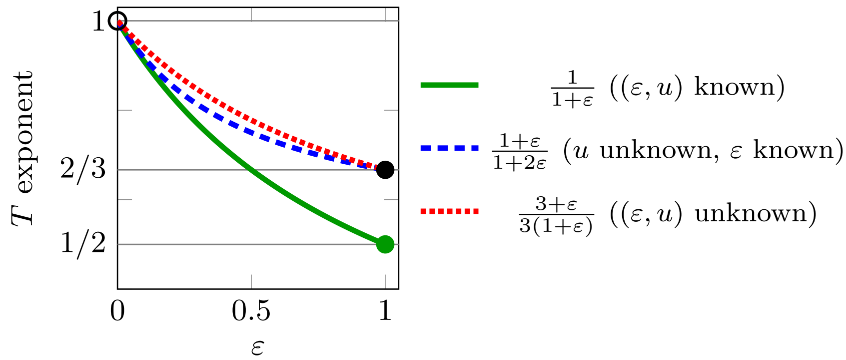

Figure 2: T exponent as a function of ϵ.

does not know the value of ϵ. The distribution-dependent guarantee follows because, on every fixed instance, the probability of committing to a suboptimal arm vanishes sufficiently fast. Asymptotically, the regret is therefore entirely due to round-robin exploration. For every fixed ϵ, Theorem 7 therefore gives the rates $\Phi_{free,\epsilon}(K,T) = \widetilde{\mathcal{O}}\big(K^{\frac{2}{3}\frac{\epsilon}{1+\epsilon}} T^{1-\frac{2}{3}\frac{\epsilon}{1+\epsilon}}\big)$ and $\Phi_{dep,\epsilon}(K,T) = \widetilde{\mathcal{O}}\big(K^{-2/3}T^{2/3}\big)$. Consequently,

$$\begin{aligned}&\Phi_{free,\epsilon}(K,T)\Phi_{dep,\epsilon}(K,T)^{\frac{\epsilon}{1+\epsilon}} \\ &\quad = \widetilde{\mathcal{O}}\big(K^{\frac{2}{3}\frac{\epsilon}{1+\epsilon}-\frac{2}{3}\frac{\epsilon}{1+\epsilon}} T^{1-\frac{2}{3}\frac{\epsilon}{1+\epsilon}+\frac{2}{3}\frac{\epsilon}{1+\epsilon}}\big) = \widetilde{\mathcal{O}}(T).\end{aligned}$$

Thus, for every fixed $\epsilon > 0$, the order-free version of `AdaR-ETC` matches the unknown-u frontier in its polynomial dependence on T. The exponents of K satisfy the corresponding trade-off on the horizon-optimal boundary.

Limits of (ϵ, u)-adaptivity. If ϵ were known, the point balancing both the T-dependence and the K-dependence on the u-adaptivity frontier would be obtained by choosing $\alpha^\star(\epsilon) = (1+\epsilon)/(1+2\epsilon)$ and $q^\star(\epsilon) = \epsilon/(1+2\epsilon)$. For fixed K, the price of not knowing ϵ is therefore the difference

$$\frac{3+\epsilon}{3(1+\epsilon)} - \frac{1+\epsilon}{1+2\epsilon} = \frac{\epsilon(1-\epsilon)}{3(1+\epsilon)(1+2\epsilon)} \geq 0. \tag{14}$$

The two exponents coincide at $\epsilon = 1$, whereas the lack of knowledge of ϵ causes a strictly positive loss in the dependence on T for every $\epsilon \in (0,1)$ (Figure 1).

When the dependence on K is also retained, the two rates are not overall comparable. Indeed,

$$\frac{2\epsilon}{3(1+\epsilon)} - \frac{\epsilon}{1+2\epsilon} = -\frac{\epsilon(1-\epsilon)}{3(1+\epsilon)(1+2\epsilon) \leq 0}.$$

Thus, the endpoint-calibrated strategy has a worse dependence on T, but a smaller distribution-free exponent in K.

This behavior is a consequence of the K-trade-off characterized in the previous sections. For every $\epsilon < 1$, we have $q^\star(\epsilon) = \epsilon/(1+2\epsilon) < 1/3$. The order-free strategy therefore explores more aggressively in K than the strategy designed with knowledge of ϵ. This additional exploration improves the distribution-free dependence on K, but worsens the dependence on K on favorable instances.

We now show that this redistribution is unavoidable. Fix an exploration strategy π that uses neither u nor ϵ. For an instance $\underline{\nu}$, define its intrinsic moment scale at order ϵ as

$$U_\epsilon(\underline{\nu}) \coloneqq \max_{i\in[K]} \left(\mathbb{E}_{X\sim\nu_i}\left[|X|^{1+\epsilon}\right]\right)^{\frac{1}{1+\epsilon}}. \tag{15}$$

The normalized regret profile of π is

$$\Phi^\pi_\epsilon(K,T) \coloneqq \sup_{\underline{\nu}:0<U_\epsilon(\underline{\nu})<+\infty} \frac{R^\pi_T(\underline{\nu})}{U_\epsilon(\underline{\nu})}. \tag{16}$$

The normalized profile is equivalent to a scale-uniform raw-moment guarantee. More precisely, for every

$B \geq 0$, $\Phi_\epsilon^\pi(K,T) \leq B$ if and only if the same strategy π satisfies $\sup_{\underline{\nu}\in\mathcal{H}_{\epsilon,u}} R_T^\pi(\underline{\nu}) \leq u^{\frac{1}{1+\epsilon}} B$ simultaneously for every $u > 0$. Indeed, every $\underline{\nu} \in \mathcal{H}_{\epsilon,u}$ satisfies $U_\epsilon(\underline{\nu}) \leq u^{1/(1+\epsilon)}$, while the reverse implication follows by setting $u = U_\epsilon(\underline{\nu})^{1+\epsilon}$.

To simplify the notation, once the strategy π is fixed, we write $\Phi_\epsilon(K,T)$ in place of $\Phi_\epsilon^\pi(K,T)$.

Theorem 8 (Pairwise lower bound for adaptation to an unknown ϵ)**.** *There exists a numerical constant $c_1 > 0$ such that, for every fixed strategy whose action rule uses neither the realized moment order nor the moment bound, $T \geq K \geq 2$, and $0 < \epsilon \leq \epsilon' \leq 1$, if $\Phi_{\epsilon'}(K,T) \leq T/4$, then*

$$\Phi_\epsilon(K,T)\Phi_{\epsilon'}(K,T)^{\frac{\epsilon}{1+\epsilon}} \geq c_1 T K^{\frac{\epsilon}{1+\epsilon}}. \tag{17}$$

Specializing Theorem 8 to $\epsilon' = 1$ gives a conditional tightness result. More precisely, among strategies retaining the endpoint guarantee $\Phi_1(K,T) \leq \widetilde{\mathcal{O}}\left(K^{1/3}T^{2/3}\right)$, the pairwise frontier forces the dependence on both K and T displayed in Theorem 7. This does not define an unconditional minimax curve over all values of ϵ: a different strategy may deliberately accept a worse guarantee at $\epsilon = 1$ to improve its performance at another moment order. Suppose that a strategy satisfies, in the non-saturated regime, $\Phi_1(K,T) \leq \widetilde{\mathcal{O}}\big(K^{1/3}T^{2/3}\big)$. Theorem 8 then gives

$$\Phi_\epsilon(K,T) \geq \widetilde{\Omega}\big(2^{-\frac{\epsilon}{1+\epsilon}} K^{\frac{2}{3}\frac{\epsilon}{1+\epsilon}} T^{1-\frac{2}{3}\frac{\epsilon}{1+\epsilon}}\big), \tag{18}$$

matching the distribution-free upper bound of Theorem 7 in both K and T, up to logarithmic factors.

Therefore, among strategies retaining the endpoint guarantee $\widetilde{\mathcal{O}}\left(K^{1/3}T^{2/3}\right)$ at $\epsilon = 1$, the distribution-free rate of the order-free version of `AdaR-ETC` is frontier-optimal, up to logarithmic factors, in its joint dependence on K and T.

The choice $q = 1/3$ can also be recovered directly from this matching requirement. Suppose that the endpoint schedule used a generic exponent q. Its endpoint distribution-free rate would have order $K^{\frac{1-q}{2}}T^{2/3}$. The pairwise lower bound would then imply, at moment order ϵ, a K-dependence of at least $K^{\frac{1+q}{2}\frac{\epsilon}{1+\epsilon}}$. On the other hand, the corresponding Median of Means upper bound would scale as $K^{\frac{\epsilon}{1+\epsilon}(1-q)}$. The two exponents coincide if and only if $(1-q)\epsilon/(1+\epsilon) = (1+q)\epsilon/(2+2\epsilon)$, which gives $q = 1/3$. Thus, the $K^{1/3}T^{2/3}$ exploration schedule is not merely a convenient endpoint choice: it is the unique polynomial schedule in this family that matches the unknown-order frontier in K and T.

Finally, pointwise sublinearity cannot be strengthened to a uniform guarantee over all moment orders.

Corollary 9 (Impossibility of uniform sublinear adaptation)**.** *There exists a numerical constant $c_2 > 0$ such that, for every strategy that uses neither u nor ϵ and every $T \geq K \geq 2$,*

$$\sup_{\epsilon\in(0,1]} \Phi_\epsilon(K,T) \geq c_2 T. \tag{19}$$

Hence, no strategy can guarantee sublinear normalized regret uniformly over $\epsilon \in (0,1]$ without further assumptions. There is no contradiction between this impossibility result and Theorem 7. The theorem fixes $\epsilon > 0$ and K before letting T grow, whereas the supremum in Corollary 9 may select a different value of ϵ for every horizon. Consistently, $\lim_{\epsilon\to 0}(3+\epsilon)/(3(1+\epsilon)) = 1$. Thus, sublinear regret is achievable for every fixed $\epsilon > 0$ and fixed K, but not uniformly over the entire range $\epsilon \in (0,1]$.

Why Calibrating to $\epsilon = 1$? The choice of calibrating `AdaR-ETC` to the finite-variance endpoint $\epsilon = 1$ may appear arbitrary, especially because the same construction can be calibrated to any design order $\bar{\epsilon} \in (0,1]$. The parameter $\bar{\epsilon}$ should not be interpreted as an estimate of the unknown true order ϵ. Rather, it selects an operating point on the adaptation frontier. Calibrating to $\bar{\epsilon} = 1$ is a natural choice because it requires no additional information and preserves the optimal guarantee on the finite-variance class. Since every admissible true order satisfies $\epsilon \leq 1$, this choice always corresponds to an *optimistic* calibration: unless $\epsilon = 1$, the algorithm overestimates the moment order and explores less than a strategy calibrated to the true class. The resulting deterioration is the price required to retain the finite-variance guarantee.

More generally, fix a calibration order $\bar{\epsilon} \in (0,1]$ and define $\bar{q} = \bar{\epsilon}/(1+2\bar{\epsilon})$ and $\bar{\beta} = (1+\bar{\epsilon})/(1+2\bar{\epsilon})$. The calibrated version of `AdaR-ETC` uses the exploration budget $\widetilde{L}_T(\bar{\epsilon}) = KB_T + \left\lceil K^{\bar{q}} T^{\bar{\beta}} \right\rceil$. It depends on $\bar{\epsilon}$, but not on ϵ or on u.

Theorem 10 (Regret of `AdaR-ETC` calibrated with $\epsilon = \bar{\epsilon}$)**.** *Fix a calibration order $\bar{\epsilon} \in (0, 1]$. Let $\epsilon \in (0, 1]$ and $u > 0$ be fixed and unknown. For every $T \geq K \geq 2$ and every instance $\underline{\nu} \in \mathcal{H}_{\epsilon,u}$, the $\bar{\epsilon}$-calibrated version of `AdaR-ETC` calibrated with $\bar{\epsilon}$ satisfies*

$$R_T(\underline{\nu}) \leq \widetilde{\mathcal{O}} \left(u^{\frac{1}{1+\epsilon}} \begin{cases} K^{\frac{\epsilon(1+\bar{\epsilon})}{(1+\epsilon)(1+2\bar{\epsilon})}} T^{\frac{1+2\bar{\epsilon}+\epsilon\bar{\epsilon}}{(1+\epsilon)(1+2\bar{\epsilon})}}, & \epsilon \leq \bar{\epsilon}, \\ K^{\frac{\bar{\epsilon}}{1+2\bar{\epsilon}}} T^{\frac{1+\bar{\epsilon}}{1+2\bar{\epsilon}}}, & \epsilon \geq \bar{\epsilon}. \end{cases} \right).$$

Moreover, for every fixed instance $\underline{\nu} \in \mathcal{H}_{\epsilon,u}$,

$$\limsup_{T \to +\infty} \frac{R_T(\underline{\nu})}{K^{-\frac{1+\bar{\epsilon}}{1+2\bar{\epsilon}}} T^{\frac{1+\bar{\epsilon}}{1+2\bar{\epsilon}}}} \leq \sum_{i:\Delta_i > 0} \Delta_i. \tag{20}$$

At the calibration order $\epsilon = \bar{\epsilon}$, the two branches coincide and give $\widetilde{\mathcal{O}}\big(u^{\frac{1}{1+\bar{\epsilon}}} K^{\frac{\bar{\epsilon}}{1+2\bar{\epsilon}}} T^{\frac{1+\bar{\epsilon}}{1+2\bar{\epsilon}}}\big)$, namely the balanced unknown-u rate associated with $\bar{\epsilon}$.

The two sides of the calibration order have different interpretations. If $\bar{\epsilon} < \epsilon$, the learner underestimates the moment order and therefore explores more than necessary. This choice is conservative: the exploration term dominates, and the regret remains at the rate associated with $\bar{\epsilon}$. If $\bar{\epsilon} > \epsilon$, the learner overestimates the moment order and explores too little for the true heavy-tailed class. The estimation term then dominates, and the regret deteriorates as the true ϵ decreases.

The pairwise lower bound in Theorem 8 shows that these two branches cannot be improved, up to logarithmic factors, while preserving the balanced guarantee at $\bar{\epsilon}$. For $\epsilon < \bar{\epsilon}$, this follows by applying the lower bound to the pair $(\epsilon, \bar{\epsilon})$; for $\epsilon > \bar{\epsilon}$, it follows by applying it to $(\bar{\epsilon}, \epsilon)$. Thus, each calibration selects a frontier-optimal profile across moment classes. In particular, calibrating to $\bar{\epsilon} = 1$ does not make the strategy simultaneously minimax-optimal at every ϵ, which is impossible. Instead, it selects the frontier-optimal profile among strategies retaining the balanced finite-variance guarantee.

7 Conclusions and Future Directions

We resolved the assumption-free rate and algorithmic components of the open problem of Genalti and Metelli (2025). We characterized the regret frontier induced by adaptation to the unknown moment bound u, provided an algorithm, `Adaptive Robust ETC`, matching it up to logarithmic factors, and extended the analysis to the case in which both u and ϵ are unknown. Two natural directions remain open. First, it would be interesting to design an anytime version of our algorithm; a doubling-trick construction should preserve the polynomial rates, at the cost of additional logarithmic factors. Second, completing the third part of the open problem requires identifying the weakest additional assumption under which the oracle rates can be recovered.

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A Proof of the Lower Bound for u-adaptive Heavy-Tailed Bandits

Theorem 4 (Existence of a trade-off)**.** *Fix $\epsilon \in (0,1]$. Consider a strategy that does not know u and admits a moment-free distribution-free rate $\Phi_{free}(K,T) = o(T)$, for every fixed $K \geq 2$. Then, any distribution-dependent rate $\Phi_{dep}(K,T)$ satisfying Definition 3 fulfills*

$$\liminf_{T\to+\infty} \frac{\Phi_{dep}(K,T)\Phi_{free}(K,T)^{\frac{1+\epsilon}{\epsilon}}}{T^{\frac{1+\epsilon}{\epsilon}}} \geq c_\epsilon, \quad \forall K \geq 2, \tag{11}$$

where $c_\epsilon > 0$ depends only on ϵ.

Proof. Let

$$\rho \coloneqq \frac{\epsilon}{1+\epsilon}, \qquad p \coloneqq \frac{1}{\rho} = \frac{1+\epsilon}{\epsilon},$$

and let

$$\Phi \coloneqq \Phi_{free}(K,T).$$

Fix $K \geq 2$ and $\Delta > 0$, and consider the deterministic instance $\underline{\nu}^{(0)}$ defined by

$$\nu_1^{(0)} = \delta_\Delta, \qquad \nu_i^{(0)} = \delta_0, \quad i \in \{2,\ldots,K\}.$$

Arm 1 is the unique optimal arm, while every arm $i \geq 2$ has gap $\Delta_i = \Delta$. Consequently,

$$R_T(\underline{\nu}^{(0)}) = \Delta \sum_{i=2}^{K} \mathbb{E}_{\underline{\nu}^{(0)}}[N_i(T)],$$

where

$$N_i(T) \coloneqq \sum_{t=1}^{T} \mathbf{1}\{I_t = i\}.$$

Since $\Phi_{free}(K,T) = o(T)$ for every fixed K, for all sufficiently large T it holds that

$$\Phi \leq \frac{2^{-\rho}}{16} T.$$

For such values of T, define

$$\beta \coloneqq \left(\frac{16\Phi}{T}\right)^p.$$

The preceding inequality ensures that $\beta \leq 1/2$.

For every suboptimal arm $i \in \{2,\ldots,K\}$, consider an alternative instance $\underline{\nu}^{(i)}$ that differs from $\underline{\nu}^{(0)}$ only in the distribution of arm i, which is replaced by

$$\nu_i^{(i)} = (1-\beta)\delta_0 + \beta\delta_{\frac{2\Delta}{\beta}}.$$

The mean of the modified arm is

$$\mu_i^{(i)} = \beta\frac{2\Delta}{\beta} = 2\Delta.$$

Therefore, arm i is the unique optimal arm under $\underline{\nu}^{(i)}$. Arm 1 has gap Δ, while every arm $j \notin \{1,i\}$ has gap 2Δ.

The $(1+\epsilon)$-moment of the modified arm is

$$\begin{aligned}\mathbb{E}_{X\sim\nu_i^{(i)}}\left[|X|^{1+\epsilon}\right] &= \beta\left(\frac{2\Delta}{\beta}\right)^{1+\epsilon} \\ &= (2\Delta)^{1+\epsilon}\beta^{-\epsilon}.\end{aligned}$$

Hence, $\underline{\nu}^{(i)}$ belongs to $\mathcal{H}_{\epsilon,u_i}$ with

$$u_i \coloneqq (2\Delta)^{1+\epsilon}\beta^{-\epsilon},$$

and

$$u_i^{\frac{1}{1+\epsilon}} = 2\Delta\beta^{-\rho}.$$

By the moment-free distribution-free guarantee,

$$\begin{aligned}R_T(\underline{\nu}^{(i)}) &\leq 2\Delta\beta^{-\rho}\Phi \\ &= 2\Delta\frac{T}{16\Phi}\Phi \\ &= \frac{\Delta T}{8},\end{aligned}$$

where we used

$$\beta^\rho = \frac{16\Phi}{T}.$$

Every pull of an arm different from i incurs regret at least Δ under $\underline{\nu}^{(i)}$. It follows that

$$\mathbb{E}_{\underline{\nu}^{(i)}}\left[T - N_i(T)\right] \leq \frac{T}{8}.$$

For ease of notation, let

$$x_i \coloneqq \mathbb{E}_{\underline{\nu}^{(0)}}[N_i(T)]$$

and

$$y_i \coloneqq \mathbb{E}_{\underline{\nu}^{(i)}}\left[T - N_i(T)\right].$$

Thus,

$$y_i \leq \frac{T}{8}.$$

Let $\mathbb{P}_0$ and $\mathbb{P}_i$ denote the distributions of the complete interaction history under $\underline{\nu}^{(0)}$ and $\underline{\nu}^{(i)}$, respectively. Since the two instances differ only on arm i, the adaptive KL decomposition gives

$$\begin{aligned}\mathrm{KL}(\mathbb{P}_0, \mathbb{P}_i) &= x_i \mathrm{KL}\left(\delta_0, (1-\beta)\delta_0 + \beta\delta_{\frac{2\Delta}{\beta}}\right) \\ &= x_i \log\left(\frac{1}{1-\beta}\right).\end{aligned}$$

Since $\beta \leq 1/2$,

$$\log\left(\frac{1}{1-\beta}\right) \leq 2\beta,$$

and therefore

$$\mathrm{KL}(\mathbb{P}_0, \mathbb{P}_i) \leq 2\beta x_i.$$

Consider the event

$$A_i \coloneqq \left\{N_i(T) > \frac{T}{2}\right\}.$$

By the definitions of x_i and y_i,

$$x_i \geq \frac{T}{2}\mathbb{P}_0(A_i)$$

and

$$y_i \geq \frac{T}{2}\mathbb{P}_i(A_i^c).$$

The Bretagnolle–Huber inequality then gives

$$\begin{aligned} x_i + y_i &\geq \frac{T}{2}\left(\mathbb{P}_0(A_i) + \mathbb{P}_i(A_i^c)\right) \\ &\geq \frac{T}{4}\exp\left(-\mathrm{KL}(\mathbb{P}_0, \mathbb{P}_i)\right) \\ &\geq \frac{T}{4}\exp(-2\beta x_i). \end{aligned}$$

Since $y_i \leq T/8$, we have

$$x_i + \frac{T}{8} \geq \frac{T}{4}\exp(-2\beta x_i).$$

We claim that

$$x_i \geq \frac{1}{32}\min\left\{T, \frac{1}{\beta}\right\}.$$

Indeed, suppose by contradiction that

$$x_i < \frac{1}{32}\min\left\{T, \frac{1}{\beta}\right\}.$$

Then

$$x_i < \frac{T}{32} \qquad \text{and} \qquad \beta x_i < \frac{1}{32}.$$

Consequently,

$$x_i + \frac{T}{8} < \frac{5T}{32},$$

whereas

$$\begin{aligned} \frac{T}{4}\exp(-2\beta x_i) &> \frac{T}{4}\exp\left(-\frac{1}{16}\right) \\ &> \frac{3T}{16} = \frac{6T}{32}. \end{aligned}$$

This is a contradiction. Therefore,

$$x_i \geq \frac{1}{32}\min\left\{T, \left(\frac{T}{16\Phi}\right)^p\right\}.$$

Summing over the $K-1$ suboptimal arms yields

$$R_T(\underline{\nu}^{(0)}) \geq \frac{K-1}{32}\Delta \min\left\{T, \left(\frac{T}{16\Phi}\right)^p\right\}.$$

It remains to remove the minimum. Let $c_{\mathrm{std},\epsilon} > 0$ be the constant appearing in Theorem 1. Since every valid moment-free distribution-free rate must satisfy the standard minimax lower bound, for all sufficiently large T,

$$\Phi \geq c_{\mathrm{std},\epsilon} K^{\rho} T^{\frac{1}{1+\epsilon}}.$$

It follows that

$$\begin{aligned} \left(\frac{T}{16\Phi}\right)^p &\leq \frac{T}{(16c_{\mathrm{std},\epsilon})^p K} \\ &\leq \frac{T}{2(16c_{\mathrm{std},\epsilon})^p}, \end{aligned}$$

where the last inequality uses $K \geq 2$.

Define

$$\kappa_\epsilon := \min\left\{1, 2(16c_{\mathrm{std},\epsilon})^p\right\}.$$

The preceding upper bound implies

$$\min\left\{T, \left(\frac{T}{16\Phi}\right)^{p}\right\} \geq \kappa_\epsilon \left(\frac{T}{16\Phi}\right)^{p}.$$

Therefore,

$$R_T(\underline{\nu}^{(0)}) \geq c_\epsilon (K-1)\Delta \left(\frac{T}{\Phi}\right)^{p},$$

where

$$c_\epsilon := \frac{\kappa_\epsilon}{32\, 16^p} = \frac{\min\left\{1, 2(16 c_{\mathrm{std},\epsilon})^{\frac{1+\epsilon}{\epsilon}}\right\}}{32\, 16^{\frac{1+\epsilon}{\epsilon}}}.$$

Thus, all constants are now explicitly specified in terms of the constant $c_{\mathrm{std},\epsilon}$ from the standard minimax lower bound.

On the baseline instance,

$$\sum_{i:\Delta_i>0} \Delta_i = (K-1)\Delta.$$

By Definition 3, for every $\eta > 0$ and all sufficiently large T,

$$R_T(\underline{\nu}^{(0)}) \leq (1+\eta)\Phi_{dep}(K,T)(K-1)\Delta.$$

Combining the upper and lower bounds and cancelling $(K-1)\Delta$ gives

$$\frac{\Phi_{dep}(K,T)\Phi_{free}(K,T)^p}{T^p} \geq \frac{c_\epsilon}{1+\eta}$$

for all sufficiently large T. Taking the inferior limit and then letting $\eta \to 0$ yields

$$\liminf_{T\to+\infty} \frac{\Phi_{dep}(K,T)\Phi_{free}(K,T)^{\frac{1+\epsilon}{\epsilon}}}{T^{\frac{1+\epsilon}{\epsilon}}} \geq c_\epsilon.$$

This concludes the proof. ■

B Proofs of the Regret Upper Bounds of `AdaR-ETC` in u-adaptive Heavy-Tailed Bandits

We first derive the Median of Means concentration inequality used in both proofs. The argument is standard and follows from the finite-moment analysis underlying robust heavy-tailed estimation (Bubeck, Cesa-Bianchi, and Lugosi, 2013).

Lemma 11 (Concentration of the Median of Means estimator)**.** *Fix $\epsilon \in (0,1]$, and let $X_1, \ldots, X_n$ be independent samples with common mean μ satisfying*

$$\max_{i\in[n]} \mathbb{E}[|X_i|^{1+\epsilon}] \leq u.$$

Let $1 \leq B \leq n$, divide the samples into B blocks of size $s = \lfloor n/B \rfloor$, and let $\widehat{\mu}^{MoM}$ be the Median of Means estimator defined from these blocks. Then,

$$\mathbb{P}\left(\left|\widehat{\mu}^{MoM} - \mu\right| > C_\epsilon^{MoM} u^{\frac{1}{1+\epsilon}} \left(\frac{B}{n}\right)^{\frac{\epsilon}{1+\epsilon}}\right) \leq \exp\left(-\frac{B}{8}\right),$$

where

$$C_\epsilon^{MoM} := 2^{1+\frac{3+\epsilon}{1+\epsilon}}.$$

Proof. Let $p = 1+\epsilon$. By Jensen's inequality,

$$|\mu|^p \leq \mathbb{E}[|X_1|^p] \leq u.$$

Therefore,

$$\mathbb{E}[|X_1-\mu|^p] \le 2^{p-1}\left(\mathbb{E}[|X_1|^p]+|\mu|^p\right) \le 2^p u.$$

Let $\overline{X}_b$ be the average of one block of size $s=\lfloor n/B \rfloor$. The von Bahr–Esseen inequality gives

$$\mathbb{E}\left[|\overline{X}_b-\mu|^p\right] \le 2^{p+1} u\, s^{1-p}.$$

By Markov's inequality,

$$\mathbb{P}\left(|\overline{X}_b-\mu| > 2^{1+\frac{3}{p}} u^{\frac{1}{p}} s^{-\frac{p-1}{p}}\right) \le \frac{1}{4}.$$

Since $s \ge n/(2B)$,

$$2^{1+\frac{3}{p}} s^{-\frac{p-1}{p}} \le 2^{2+\frac{2}{p}} \left(\frac{B}{n}\right)^{\frac{p-1}{p}} = C_\epsilon^{MoM} \left(\frac{B}{n}\right)^{\frac{\epsilon}{1+\epsilon}}.$$

If the Median of Means estimator deviates from μ by more than the displayed radius, at least half of the block averages must be bad. The blocks are independent, and each is bad with probability at most 1/4. Hoeffding's inequality therefore yields

$$\mathbb{P}\left(\left|\widehat{\mu}^{MoM}-\mu\right| > C_\epsilon^{MoM} u^{\frac{1}{1+\epsilon}} \left(\frac{B}{n}\right)^{\frac{\epsilon}{1+\epsilon}}\right) \le \exp\left(-\frac{B}{8}\right).$$

■

Theorem 5 (Distribution-free regret of `AdaR-ETC`)**.** *Let $\epsilon \in (0,1]$ be fixed and known. Let $\alpha \in [(1+\epsilon)/(1+2\epsilon), 1)$ and $q \in [0, \epsilon/(1+2\epsilon)]$. For every $u>0$ and every instance $\underline{\nu} \in \mathcal{H}_{\epsilon,u}$, `AdaR-ETC` satisfies*

$$R_T^{\texttt{AdaR-ETC}}(\underline{\nu}) \le \widetilde{\mathcal{O}}\big(u^{\frac{1}{1+\epsilon}} K^{\frac{\epsilon}{1+\epsilon}(1-q)} T^\alpha\big),$$

where $\widetilde{\mathcal{O}}$ hides polylogarithmic terms in T and constants depending only on ϵ, α, and q.

Proof. Let

$$\rho_\epsilon := \frac{\epsilon}{1+\epsilon}, \qquad r_q := \rho_\epsilon(1-q),$$

and recall that

$$\beta_\alpha = \frac{1-\alpha}{\rho_\epsilon}.$$

For ease of notation, define

$$V := u^{\frac{1}{1+\epsilon}}, \qquad M_T := K^q T^{\beta_\alpha}, \qquad H_T := K^{r_q} T^\alpha.$$

By Jensen's inequality, $|\mu_i| \le V$ for every arm. Therefore, every suboptimality gap is bounded by

$$\Delta_i \le 2V.$$

The restrictions on q and α imply

$$q \le \rho_\epsilon(1-q) = r_q$$

and

$$\beta_\alpha \le \alpha.$$

Consequently,

$$M_T \le H_T.$$

Moreover, $q \le \epsilon/(1+2\epsilon)$ implies $r_q \ge \epsilon/(1+2\epsilon)$, while $\alpha \ge (1+\epsilon)/(1+2\epsilon)$. Hence,

$$\alpha \ge 1 - r_q.$$

We now bound the exploration length. If $K \le T$, then

$$K = K^{r_q} K^{1-r_q} \le K^{r_q} T^{1-r_q} \le H_T,$$

and therefore

$$L_T \le KB_T + M_T + 1 \le (B_T + 2)H_T.$$

If $K > T$, then $\widetilde{L}_T \ge KB_T > T$, so that $L_T = T$. Since $r_q + \alpha \ge 1$, we also have

$$L_T = T \le K^{r_q} T^{\alpha} = H_T.$$

Thus, in all cases,

$$L_T \le (B_T + 2)H_T.$$

If $L_T = T$, the algorithm only explores, and the regret satisfies

$$R_T(\underline{\nu}) \le 2VL_T \le 2V(B_T + 2)H_T.$$

We may therefore assume that $L_T < T$. In this case, $L_T = \widetilde{L}_T$, and every arm receives at least

$$n_i \ge \left\lfloor \frac{L_T}{K} \right\rfloor \ge B_T$$

samples. Moreover, writing $M_T = K^q T^{\beta_\alpha}$ and using $B_T \ge 1$,

$$\begin{aligned} n_i &\ge \left\lfloor \frac{KB_T + \lceil M_T \rceil}{K} \right\rfloor \\ &= B_T + \left\lfloor \frac{\lceil M_T \rceil}{K} \right\rfloor \\ &\ge B_T + \frac{M_T}{K} - 1 \ge K^{q-1} T^{\beta_\alpha}. \end{aligned}$$

Let $\mathcal{E}_T$ be the event on which, simultaneously for every $i \in [K]$,

$$\left|\widehat{\mu}_i^{MoM} - \mu_i\right| \le C_\epsilon^{MoM} V \left(\frac{B_T}{n_i}\right)^{\rho_\epsilon}.$$

By Lemma 11 and a union bound,

$$\mathbb{P}(\mathcal{E}_T^c) \le K \exp\left(-\frac{B_T}{8}\right).$$

Since $B_T \ge 8\log(KT^3)$,

$$\mathbb{P}(\mathcal{E}_T^c) \le \frac{1}{T^3}.$$

On $\mathcal{E}_T$, every estimate has error at most

$$r_T \coloneqq C_\epsilon^{MoM} V B_T^{\rho_\epsilon} K^{r_q} T^{-\rho_\epsilon \beta_\alpha}.$$

Since $\widehat{I}^*$ maximizes the estimated mean, its suboptimality gap satisfies

$$\Delta_{\widehat{I}^*} \le 2r_T.$$

The expected regret can therefore be bounded as

$$R_T(\underline{\nu}) \le 2VL_T + 2r_T T + 2VT\mathbb{P}(\mathcal{E}_T^c).$$

Using

$$1 - \rho_\epsilon \beta_\alpha = \alpha,$$

we obtain

$$2r_T T = 2C_\epsilon^{MoM} V B_T^{\rho_\epsilon} H_T.$$

Combining the previous bounds gives

$$R_T(\underline{\nu}) \le 2V(B_T + 2)H_T + 2C_\epsilon^{MoM} V B_T^{\rho_\epsilon} H_T + \frac{2V}{T^2}.$$

Since $B_T \geq 1$, $\rho_\epsilon \leq 1$, and $H_T \geq 1$, it follows that

$$R_T(\underline{\nu}) \leq \left(2C_\epsilon^{MoM} + 8\right) B_T V H_T.$$

Recalling the definitions of V and H_T, we conclude that

$$R_T(\underline{\nu}) \leq \left(2^{2+\frac{3+\epsilon}{1+\epsilon}} + 8\right) B_T u^{\frac{1}{1+\epsilon}} K^{\frac{\epsilon}{1+\epsilon}(1-q)} T^\alpha.$$

Since $B_T = \lceil 8\log(KT^3)\rceil$, the stated $\widetilde{\mathcal{O}}$ guarantee follows. ■

Theorem 6 (Distribution-dependent regret of `AdaR-ETC`)**.** *Let $\epsilon \in (0,1]$ be fixed and known. Let $\alpha \in [(1+\epsilon)/(1+2\epsilon), 1)$ and $q \in [0, \epsilon/(1+2\epsilon)]$. For every fixed instance $\underline{\nu} \in \mathcal{H}_{\epsilon,u}$, `AdaR-ETC` satisfies*

$$\limsup_{T\to+\infty} \frac{R_T^{\texttt{AdaR-ETC}}(\underline{\nu})}{K^{q-1}T^{\beta_\alpha}} \leq \sum_{i:\Delta_i>0} \Delta_i.$$

Proof. Let

$$\rho_\epsilon \coloneqq \frac{\epsilon}{1+\epsilon}, \qquad V \coloneqq u^{\frac{1}{1+\epsilon}}.$$

If every arm is optimal, the regret is identically zero and the claim is immediate. We may therefore assume that the instance contains at least one suboptimal arm, and define

$$\Delta_{\min} \coloneqq \min_{i:\Delta_i>0} \Delta_i.$$

Since $\alpha < 1$, we have $\beta_\alpha > 0$. Moreover,

$$\beta_\alpha \leq \frac{1+\epsilon}{1+2\epsilon} < 1.$$

For every fixed K, it follows that

$$KB_T + K^q T^{\beta_\alpha} = o(T).$$

Thus, for all sufficiently large T, we have $L_T = \widetilde{L}_T < T$.

As in the proof of Theorem 5, every arm receives at least

$$n_i \geq K^{q-1}T^{\beta_\alpha}$$

exploration samples. Define

$$r_T \coloneqq C_\epsilon^{MoM} V \left(\frac{B_T K^{1-q}}{T^{\beta_\alpha}}\right)^{\rho_\epsilon}.$$

Since K and the instance are fixed, B_T is logarithmic in T and $\beta_\alpha > 0$, so that

$$\lim_{T\to+\infty} r_T = 0.$$

Consequently, for all sufficiently large T,

$$2r_T < \Delta_{\min}.$$

Let $\mathcal{E}_T$ be the event on which all the MoM estimates differ from their respective means by at most r_T. Lemma 11 and the choice $B_T \geq 8\log(KT^3)$ give

$$\mathbb{P}(\mathcal{E}_T^c) \leq \frac{1}{T^3}.$$

On $\mathcal{E}_T$, no suboptimal arm can maximize the estimated mean. Therefore, the arm selected during the commitment phase is optimal.

During the round-robin exploration phase, every arm is selected at most

$$\left\lceil \frac{L_T}{K} \right\rceil \leq B_T + K^{q-1}T^{\beta_\alpha} + 2$$

times. Hence, the exploration regret is bounded by

$$\big(B_T + K^{q-1}T^{\beta_\alpha} + 2\big) \sum_{i:\Delta_i>0} \Delta_i.$$

The commitment phase incurs no regret on $\mathcal{E}_T$. On $\mathcal{E}_T^c$, its regret is at most $2VT$. We thus obtain, for every sufficiently large T,

$$R_T(\underline{\nu}) \leq \big(B_T + K^{q-1}T^{\beta_\alpha} + 2\big) \sum_{i:\Delta_i>0} \Delta_i + \frac{2V}{T^2}.$$

Dividing by $K^{q-1}T^{\beta_\alpha}$ gives

$$\frac{R_T(\underline{\nu})}{K^{q-1}T^{\beta_\alpha}} \leq \left(1 + \frac{B_T+2}{K^{q-1}T^{\beta_\alpha}}\right) \sum_{i:\Delta_i>0} \Delta_i + \frac{2V}{K^{q-1}T^{\beta_\alpha+2}}.$$

For every fixed K,

$$\lim_{T\to+\infty} \frac{B_T+2}{K^{q-1}T^{\beta_\alpha}} = 0$$

and

$$\lim_{T\to+\infty} \frac{2V}{K^{q-1}T^{\beta_\alpha+2}} = 0.$$

Taking the superior limit proves

$$\limsup_{T\to+\infty} \frac{R_T(\underline{\nu})}{K^{q-1}T^{\beta_\alpha}} \leq \sum_{i:\Delta_i>0} \Delta_i.$$

■

C Proofs for (ϵ, u)-adaptivity

Let the constant from Lemma 11 be

$$C_\epsilon^{MoM} := 2^{1+\frac{3+\epsilon}{1+\epsilon}}$$

Theorem 7 (Regret of `AdaR-ETC` calibrated with $\epsilon = 1$)**.** *Let $\epsilon \in (0,1]$ and $u > 0$ be fixed. For every $T \geq K \geq 2$ and every instance $\underline{\nu} \in \mathcal{H}_{\epsilon,u}$, the order-free version of `AdaR-ETC` calibrated with $\epsilon = 1$ satisfies*

$$R_T^{\texttt{AdaR-ETC}}(\underline{\nu}) \leq \widetilde{\mathcal{O}}\left(u^{\frac{1}{1+\epsilon}} K^{\frac{2\epsilon}{3(1+\epsilon)}} T^{\frac{3+\epsilon}{3(1+\epsilon)}}\right). \tag{12}$$

where $\widetilde{\mathcal{O}}$ hides factors at most polylogarithmic in K and T and constants depending on ϵ. Moreover, for every fixed instance $\underline{\nu} \in \mathcal{H}_{\epsilon,u}$,

$$\limsup_{T\to+\infty} \frac{R_T^{\texttt{AdaR-ETC}}(\underline{\nu})}{K^{-2/3}T^{2/3}} \leq \sum_{i:\Delta_i>0} \Delta_i. \tag{13}$$

Proof. Let

$$\rho_\epsilon := \frac{\epsilon}{1+\epsilon}, \qquad V := u^{\frac{1}{1+\epsilon}},$$

and define

$$M_T := K^{1/3}T^{2/3}, \qquad H_T := K^{\frac{2\rho_\epsilon}{3}} T^{1-\frac{2\rho_\epsilon}{3}}.$$

By Jensen's inequality, $|\mu_i| \leq V$ for every arm, and hence every suboptimality gap satisfies $\Delta_i \leq 2V$.

Since $\rho_\epsilon \leq 1/2$ and $T \geq K$, we have

$$\frac{M_T}{H_T} = \left(\frac{K}{T}\right)^{\frac{1-2\rho_\epsilon}{3}} \leq 1$$

and

$$\frac{K}{H_T} = \left(\frac{K}{T}\right)^{1-\frac{2\rho_\epsilon}{3}} \leq 1.$$

It follows that

$$L_T \leq KB_T + M_T + 1 \leq (B_T + 2)H_T.$$

If $L_T = T$, the algorithm performs only round-robin exploration, and therefore

$$R_T(\underline{\nu}) \leq 2VL_T \leq 2V(B_T + 2)H_T.$$

We may thus assume that $L_T < T$. In this case, $L_T = \widetilde{L}_T$, $M_T = K^{1/3}T^{2/3}$, and the number n_i of samples collected from each arm satisfies

$$\begin{aligned} n_i &\geq \left\lfloor \frac{KB_T + \lceil M_T \rceil}{K} \right\rfloor \\ &= B_T + \left\lfloor \frac{\lceil M_T \rceil}{K} \right\rfloor \\ &\geq B_T + \frac{M_T}{K} - 1 \geq K^{-2/3}T^{2/3}. \end{aligned}$$

In particular, $n_i \geq B_T$, so that the Median of Means estimator is well defined.

Let $\mathcal{E}_T$ be the event on which, simultaneously for every $i \in [K]$,

$$\left|\widehat{\mu}_i^{MoM} - \mu_i\right| \leq C_\epsilon^{MoM} V \left(\frac{B_T}{n_i}\right)^{\rho_\epsilon}.$$

By Lemma 11 and a union bound,

$$\mathbb{P}(\mathcal{E}_T^c) \leq K \exp\left(-\frac{B_T}{8}\right) \leq \frac{1}{T^3}.$$

On $\mathcal{E}_T$, the estimation error of every arm is at most

$$r_T \coloneqq C_\epsilon^{MoM} V B_T^{\rho_\epsilon} K^{\frac{2\rho_\epsilon}{3}} T^{-\frac{2\rho_\epsilon}{3}}.$$

Since $\widehat{I}^*$ maximizes the estimated mean, its gap satisfies

$$\Delta_{\widehat{I}^*} \leq 2r_T.$$

The expected regret is therefore bounded by

$$R_T(\underline{\nu}) \leq 2VL_T + 2r_T T + 2VT\mathbb{P}(\mathcal{E}_T^c).$$

Using the preceding bounds, we obtain

$$\begin{aligned} R_T(\underline{\nu}) &\leq 2V(B_T + 2)H_T + 2C_\epsilon^{MoM} V B_T^{\rho_\epsilon} H_T + \frac{2V}{T^2} \\ &\leq \left(8 + 2C_\epsilon^{MoM}\right) B_T V H_T, \end{aligned}$$

where we used $B_T \geq 1$, $\rho_\epsilon \leq 1$, and $H_T \geq 1$.

Recalling the definitions of V and H_T, we conclude that

$$R_T(\underline{\nu}) \leq \left(8 + 2^{2+\frac{3+\epsilon}{1+\epsilon}}\right) B_T u^{\frac{1}{1+\epsilon}} K^{\frac{2\epsilon}{3(1+\epsilon)}} T^{\frac{3+\epsilon}{3(1+\epsilon)}}.$$

This proves the distribution-free guarantee.

We now prove the distribution-dependent claim. If every arm is optimal, then the regret is identically zero. Otherwise, define

$$\Delta_{\min} \coloneqq \min_{i:\Delta_i > 0} \Delta_i.$$

For every fixed K,

$$KB_T + K^{1/3}T^{2/3} = o(T),$$

and hence $L_T = \widetilde{L}_T < T$ for all sufficiently large T.

On the event $\mathcal{E}_T$, every estimate has error at most r_T. Since the instance and K are fixed,

$$\lim_{T\to+\infty} r_T = 0.$$

Consequently, for all sufficiently large T,

$$2r_T < \Delta_{\min},$$

and the arm selected during the commitment phase is optimal.

During round-robin exploration, every arm is pulled at most

$$\left\lceil \frac{L_T}{K} \right\rceil \le B_T + K^{-2/3}T^{2/3} + 2$$

times. The exploration regret is thus at most

$$\left(B_T + K^{-2/3}T^{2/3} + 2\right) \sum_{i:\Delta_i>0} \Delta_i.$$

The commitment phase incurs no regret on $\mathcal{E}_T$ and at most $2VT$ regret on $\mathcal{E}_T^c$. Therefore, for all sufficiently large T,

$$R_T(\underline{\nu}) \le \left(B_T + K^{-2/3}T^{2/3} + 2\right) \sum_{i:\Delta_i>0} \Delta_i + \frac{2V}{T^2}.$$

Dividing by $K^{-2/3}T^{2/3}$ and taking the superior limit gives

$$\limsup_{T\to+\infty} \frac{R_T(\underline{\nu})}{K^{-2/3}T^{2/3}} \le \sum_{i:\Delta_i>0} \Delta_i,$$

because $B_T = \mathcal{O}(\log T)$ for every fixed K. ■

Theorem 8 (Pairwise lower bound for adaptation to an unknown ϵ)**.** *There exists a numerical constant $c_1 > 0$ such that, for every fixed strategy whose action rule uses neither the realized moment order nor the moment bound, $T \ge K \ge 2$, and $0 < \epsilon \le \epsilon' \le 1$, if $\Phi_{\epsilon'}(K,T) \le T/4$, then*

$$\Phi_\epsilon(K,T)\Phi_{\epsilon'}(K,T)^{\frac{\epsilon}{1+\epsilon}} \ge c_1 T K^{\frac{\epsilon}{1+\epsilon}}. \tag{17}$$

Proof. Let

$$a := \frac{\epsilon}{1+\epsilon}, \qquad \Phi' := \Phi_{\epsilon'}(K,T).$$

We first prove that

$$\Phi' \ge \frac{K-1}{2}.$$

Fix $\Delta > 0$ and consider the all-zero reference process, in which every arm deterministically returns zero. For a fixed realization of the strategy's internal randomness, let

$$\tau_j = \inf\{t \in [T] : I_t = j\},$$

with $\tau_j = +\infty$ if arm j is never selected, and set

$$\sigma_j = (\tau_j - 1) \wedge T.$$

Let m be the number of arms selected at least once, and denote their ordered first-visit times by

$$r_1 < r_2 < \cdots < r_m.$$

Since at least ℓ distinct arms must have been visited by round r_ℓ, we have $r_\ell \ge \ell$. Moreover, every unvisited arm contributes T to the sum of the σ_j. Since $T \geq K$,

$$\begin{aligned}\sum_{j=1}^{K} \sigma_j &\geq \sum_{\ell=1}^{m} (r_\ell - 1) + (K-m)T \\ &\geq \frac{m(m-1)}{2} + (K-m)K \\ &\geq \frac{K(K-1)}{2}.\end{aligned}$$

Taking expectation with respect to the internal randomness of the strategy, there exists an arm $j \in [K]$ such that

$$\mathbb{E}_0[\sigma_j] \geq \frac{K-1}{2}.$$

Now consider the deterministic instance in which arm j always returns Δ and every other arm always returns zero. Until arm j is first selected, the observed history coincides with the all-zero reference process. Therefore, the regret on this instance is at least

$$\Delta \mathbb{E}_0[\sigma_j].$$

Its intrinsic moment scale at order ϵ' equals Δ, and hence

$$\Phi' \geq \frac{\Delta \mathbb{E}_0[\sigma_j]}{\Delta} \geq \frac{K-1}{2}.$$

Fix $\Delta > 0$ and consider the baseline instance $\underline{\nu}^{(0)}$ defined by

$$\nu_1^{(0)} = \delta_\Delta, \qquad \nu_i^{(0)} = \delta_0, \quad i \in \{2, \ldots, K\}.$$

Its intrinsic scale at order ϵ' is Δ. Hence,

$$R_T(\underline{\nu}^{(0)}) = \Delta \sum_{i=2}^{K} \mathbb{E}_0[N_i(T)] \leq \Delta \Phi'.$$

It follows that

$$\mathbb{E}_0[N_1(T)] \geq T - \Phi' \geq \frac{3T}{4}.$$

Moreover, there exists an arm $j \in \{2, \ldots, K\}$ such that

$$\mathbb{E}_0[N_j(T)] \leq \frac{\Phi'}{K-1}.$$

Define

$$\beta := \frac{K-1}{128\Phi'}.$$

The preliminary lower bound on Φ' ensures that $\beta \leq 1/64$. Construct an alternative instance $\underline{\nu}^{(j)}$ by replacing only arm j with

$$\nu_j^{(j)} = (1-\beta)\delta_0 + \beta \delta_{\frac{2\Delta}{\beta}}.$$

The mean of the modified arm is 2Δ, so arm j is optimal and arm 1 has gap Δ.

The intrinsic scale of the alternative instance at order ϵ is

$$\begin{aligned}U_\epsilon(\underline{\nu}^{(j)}) &= \left(\beta \left(\frac{2\Delta}{\beta}\right)^{1+\epsilon}\right)^{\frac{1}{1+\epsilon}} \\ &= 2\Delta \beta^{-\frac{\epsilon}{1+\epsilon}}.\end{aligned}$$

Consequently,

$$R_T(\underline{\nu}^{(j)}) \leq 2\Delta \beta^{-\frac{\epsilon}{1+\epsilon}} \Phi_\epsilon(K, T).$$

Let $\mathbb{P}_0$ and $\mathbb{P}_j$ denote the laws of the complete interaction history under the baseline and alternative instances. By the adaptive KL decomposition,

$$\begin{aligned}\mathrm{KL}(\mathbb{P}_0, \mathbb{P}_j) &= \mathbb{E}_0[N_j(T)] \log\left(\frac{1}{1-\beta}\right) \\ &\le 2\beta \frac{\Phi'}{K-1} = \frac{1}{64}.\end{aligned}$$

Since $N_1(T)/T \in [0,1]$, Pinsker's inequality gives

$$\begin{aligned}\left|\frac{\mathbb{E}_0[N_1(T)]}{T} - \frac{\mathbb{E}_j[N_1(T)]}{T}\right| &\le \sqrt{\frac{\mathrm{KL}(\mathbb{P}_0, \mathbb{P}_j)}{2}} \\ &\le \frac{1}{8\sqrt{2}} \le \frac{1}{8}.\end{aligned}$$

It follows that

$$\mathbb{E}_j[N_1(T)] \ge \frac{5T}{8}.$$

Every pull of arm 1 incurs regret Δ under the alternative instance, and hence

$$R_T(\underline{\nu}^{(j)}) \ge \frac{5\Delta T}{8}.$$

Combining the upper and lower bounds on the alternative regret yields

$$\Phi_\epsilon(K,T) \ge \frac{5}{16} T\beta^a.$$

Therefore,

$$\begin{aligned}\Phi_\epsilon(K,T)\Phi_{\epsilon'}(K,T)^a &\ge \frac{5}{16} T \left(\frac{K-1}{128}\right)^a \\ &\ge \frac{5}{16\sqrt{128}} T(K-1)^a \\ &\ge \frac{5}{16\sqrt{256}} TK^a\end{aligned}$$

where the last two inequalities use $a \le 1/2$. The claim follows with

$$c_1 := \frac{5}{16\sqrt{256}} = \frac{5}{256\sqrt{2}}.$$

■

Corollary 9 (Impossibility of uniform sublinear adaptation)**.** *There exists a numerical constant $c_2 > 0$ such that, for every strategy that uses neither u nor ϵ and every $T \ge K \ge 2$,*

$$\sup_{\epsilon \in (0,1]} \Phi_\epsilon(K,T) \ge c_2 T. \tag{19}$$

Proof. Let

$$c_2 := \frac{5}{16\sqrt{128}}.$$

If

$$\Phi_1(K,T) > \frac{T}{4},$$

then

$$\sup_{\epsilon \in (0,1]} \Phi_\epsilon(K,T) \ge \Phi_1(K,T) > \frac{T}{4} \ge c_2 T.$$

Suppose instead that $\Phi_1(K,T) \le T/4$. For every $\epsilon \in (0,1]$, Theorem 8 with $\epsilon' = 1$ gives

$$\begin{aligned}\Phi_\epsilon(K,T) &\ge c_2 T (K-1)^{\frac{\epsilon}{1+\epsilon}} \Phi_1(K,T)^{-\frac{\epsilon}{1+\epsilon}} \\ &\ge c_2 T \left(\frac{4(K-1)}{T}\right)^{\frac{\epsilon}{1+\epsilon}}.\end{aligned}$$

Letting ϵ decrease to zero, the last multiplicative factor converges to one. Therefore,

$$\sup_{\epsilon \in (0,1]} \Phi_\epsilon(K,T) \ge c_2 T.$$

This proves the claim. ■

Theorem 10 (Regret of `AdaR-ETC` calibrated with $\epsilon = \bar{\epsilon}$)**.** *Fix a calibration order $\bar{\epsilon} \in (0,1]$. Let $\epsilon \in (0,1]$ and $u > 0$ be fixed and unknown. For every $T \ge K \ge 2$ and every instance $\underline{\nu} \in \mathcal{H}_{\epsilon,u}$, the $\bar{\epsilon}$-calibrated version of `AdaR-ETC` calibrated with $\bar{\epsilon}$ satisfies*

$$R_T(\underline{\nu}) \le \widetilde{\mathcal{O}}\left(u^{\frac{1}{1+\epsilon}} \begin{cases} K^{\frac{\epsilon(1+\bar{\epsilon})}{(1+\epsilon)(1+2\bar{\epsilon})}} T^{\frac{1+2\bar{\epsilon}+\epsilon\bar{\epsilon}}{(1+\epsilon)(1+2\bar{\epsilon})}}, & \epsilon \le \bar{\epsilon}, \\ K^{\frac{\bar{\epsilon}}{1+2\bar{\epsilon}}} T^{\frac{1+\bar{\epsilon}}{1+2\bar{\epsilon}}}, & \epsilon \ge \bar{\epsilon}. \end{cases} \right).$$

Moreover, for every fixed instance $\underline{\nu} \in \mathcal{H}_{\epsilon,u}$,

$$\limsup_{T \to +\infty} \frac{R_T(\underline{\nu})}{K^{-\frac{1+\bar{\epsilon}}{1+2\bar{\epsilon}}} T^{\frac{1+\bar{\epsilon}}{1+2\bar{\epsilon}}}} \le \sum_{i:\Delta_i > 0} \Delta_i. \tag{20}$$

Proof. Let

$$\rho \coloneqq \frac{\epsilon}{1+\epsilon}, \qquad \bar{\rho} \coloneqq \frac{\bar{\epsilon}}{1+\bar{\epsilon}}, \qquad V \coloneqq u^{\frac{1}{1+\epsilon}}.$$

The calibration parameters can equivalently be written as

$$\bar{q} = \frac{\bar{\rho}}{1+\bar{\rho}}, \qquad \bar{\beta} = \frac{1}{1+\bar{\rho}}.$$

In particular, $1 - \bar{q} = \bar{\beta}$ and $\bar{q} + \bar{\beta} = 1$.

Define

$$M_T \coloneqq K^{\bar{q}} T^{\bar{\beta}}, \qquad H_T \coloneqq K^{\rho\bar{\beta}} T^{1-\rho\bar{\beta}}, \qquad G_T \coloneqq \max\{M_T, H_T\}.$$

Since $T \ge K$ and $\bar{q} + \bar{\beta} = 1$, we have $M_T \ge K$. Therefore,

$$L_T \le K B_T + M_T + 1 \le (B_T + 2) G_T.$$

If $L_T = T$, Jensen's inequality gives $|\mu_i| \le V$ and hence $\Delta_i \le 2V$ for every arm. Thus,

$$R_T(\underline{\nu}) \le 2V L_T \le 2V(B_T + 2) G_T.$$

Suppose now that $L_T < T$. In this case, $L_T = \widetilde{L}_T(\bar{\epsilon})$, and every arm receives at least

$$\begin{aligned} n_i &\ge \left\lfloor \frac{K B_T + \lceil M_T \rceil}{K} \right\rfloor \\ &\ge B_T + \frac{M_T}{K} - 1 \ge \frac{M_T}{K} \end{aligned}$$

exploration samples.

Let $\mathcal{E}_T$ be the event on which all the Median of Means estimates satisfy the concentration bound of Lemma 11. Since $B_T = \lceil 8 \log(KT^3) \rceil$, a union bound gives

$$\mathbb{P}(\mathcal{E}_T^c) \le K \exp\left(-\frac{B_T}{8}\right) \le \frac{1}{T^3}.$$

On $\mathcal{E}_T$, every estimate has error at most

$$\begin{aligned} r_T &\coloneqq C_\epsilon^{MoM} V \left(\frac{B_T}{n_i}\right)^\rho \\ &\leq C_\epsilon^{MoM} V B_T^\rho K^{\rho(1-\bar{q})} T^{-\rho\bar{\beta}} \\ &= C_\epsilon^{MoM} V B_T^\rho K^{\rho\bar{\beta}} T^{-\rho\bar{\beta}}. \end{aligned}$$

Since the committed arm maximizes the estimated mean, its gap is at most $2r_T$. Consequently,

$$\begin{aligned} R_T(\underline{\nu}) &\leq 2VL_T + 2r_T T + 2VT\mathbb{P}(\mathcal{E}_T^c) \\ &\leq 2V(B_T+2)G_T + 2C_\epsilon^{MoM} V B_T^\rho H_T + \frac{2V}{T^2}. \end{aligned}$$

Using $B_T \geq 1$, $\rho \leq 1$, and $G_T \geq 1$, we obtain the explicit bound

$$R_T(\underline{\nu}) \leq \left(8 + 2C_\epsilon^{MoM}\right) B_T V G_T. \tag{21}$$

It remains to identify which term defines G_T. We have

$$\frac{M_T}{H_T} = \left(\frac{K}{T}\right)^{\frac{\bar{\rho}-\rho}{1+\bar{\rho}}}.$$

Since $T \geq K$, if $\epsilon \leq \bar{\epsilon}$, then $\rho \leq \bar{\rho}$ and $M_T \leq H_T$. Equation (21) therefore gives

$$R_T(\underline{\nu}) \leq \left(8 + 2C_\epsilon^{MoM}\right) B_T V K^{\frac{\epsilon(1+\bar{\epsilon})}{(1+\epsilon)(1+2\bar{\epsilon})}} T^{1-\frac{\epsilon(1+\bar{\epsilon})}{(1+\epsilon)(1+2\bar{\epsilon})}}.$$

If instead $\epsilon \geq \bar{\epsilon}$, then $\rho \geq \bar{\rho}$ and $M_T \geq H_T$, yielding

$$R_T(\underline{\nu}) \leq \left(8 + 2C_\epsilon^{MoM}\right) B_T V K^{\frac{\bar{\epsilon}}{1+2\bar{\epsilon}}} T^{\frac{1+\bar{\epsilon}}{1+2\bar{\epsilon}}}.$$

This proves the distribution-free guarantee.

We finally prove the distribution-dependent guarantee. If every arm is optimal, the claim is immediate. Otherwise, define

$$\Delta_{\min} \coloneqq \min_{i:\Delta_i>0} \Delta_i.$$

For every fixed K and fixed $\bar{\epsilon} > 0$,

$$KB_T + K^{\bar{q}} T^{\bar{\beta}} = o(T),$$

so $L_T < T$ for all sufficiently large T. Moreover, the radius r_T converges to zero for every fixed true $\epsilon > 0$. Hence, for all sufficiently large T, $2r_T < \Delta_{\min}$ and the committed arm is optimal on $\mathcal{E}_T$.

During round-robin exploration, every arm is selected at most

$$B_T + K^{\bar{q}-1} T^{\bar{\beta}} + 2$$

times. Therefore,

$$R_T(\underline{\nu}) \leq \left(B_T + K^{\bar{q}-1} T^{\bar{\beta}} + 2\right) \sum_{i:\Delta_i>0} \Delta_i + \frac{2V}{T^2}.$$

Dividing by $K^{\bar{q}-1} T^{\bar{\beta}}$ and taking the superior limit gives

$$\limsup_{T\to+\infty} \frac{R_T(\underline{\nu})}{K^{\bar{q}-1} T^{\bar{\beta}}} \leq \sum_{i:\Delta_i>0} \Delta_i.$$

Since $\bar{q} - 1 = -(1+\bar{\epsilon})/(1+2\bar{\epsilon})$, this is exactly Equation (20). ∎